THE SIPRO PRE-PLE SET II 2022

INTEGRATED SCIENCE

Time Allowed: *2 Hours 15 Minutes*

	Random No.						Personal No.		
Index No.									

Candidate's Name: _____

Candidate's signature: _____

School Random No. _____

District ID: _____

READ THE FOLLOWING INSTRUCTIONS CAREFULLY:

1. This paper has two sections: **A** and **B**.
2. Section **A** has **40** questions (**40 Marks**).
3. Section **B** has **15** questions (**60 Marks**).
4. Attempt all questions in both sections. All answers to both sections **A** and **B**. must be written in the spaces provided.
5. All answers must be written in blue or black ball point pens or ***ink***. Only diagrams and graph work must be done in ***pencil***.
6. Unnecessary ***alteration*** of work will lead to loss of marks.
7. Any ***handwriting*** that cannot be easily read may lead to loss of marks.
8. Do not fill anything in the boxes indicated: "FOR EXAMINER'S USE ONLY"

For Examiner's Use Only;

PAGES	MARKS	INITIALS
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Page 7		
Page 8		
Page 9		
Total		

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SECTION A: 40 MARKS

*Questions **1** to **40** carry **one** mark each.*

1. Name the vector responsible for the spread of both trachoma and diarrhoea.

2. Mention the component of air which supports rusting of metallic objects.

3. Write down **one** function of the ovary in the human reproductive system.

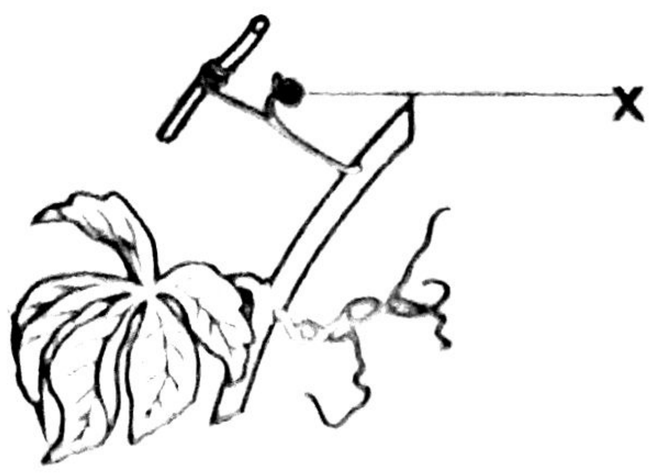
4. Give **one** cause of anti-social behaviour in a school community.

5. Why does sugar disappear when put in water?

6. State the importance of a strip cup to a dairy farmer.

7. Name the body sense organ that enables us to detect tasty food.

The diagram below shows a method of climbing. *Study and use it to answer questions 8 and 9.*



8. Name the part marked with letter **X** on the diagram above.

9. How important is the part you have named in No. 8 above to the plant?

10 What safety measure can be taken by electrical engineers to control short circuits in an installation?

11. How are red blood cells important in the human body?

12. Which property of magnets is seen when a magnet is dropped in iron filings and they settle mostly at the ends?

13. Name the dangerous gas found in tobacco smoke.

14. How best can cattle farmers improve the quality of their local breeds of cattle?

15. In the space provided below, draw a beak of a nectar sucking bird.

16. What name is given to the smallest unit of an organism?

17. Apart from brushing and rinsing, name any other activity done by people to promote oral health.

18. Mention **one** way pineapples can be propagated.

19. Give the reason why people burning bricks cover them with mud.

20. Why is a person suffering from gonorrhoea advised to go with his/her sexual partner for treatment?

21. Give **one** traditional method of preserving food.

22. Name the reproductive cell shown in the diagram below.



23. How are mammals similar to birds in terms of reproduction?

24. State the type of bee with a pollen basket.

25. Why is a spider regarded as an arachnid?

26. Give any **one** sign of brain damage caused by alcoholism to an individual

27. Why is the trachea made up of rings of cartilage?

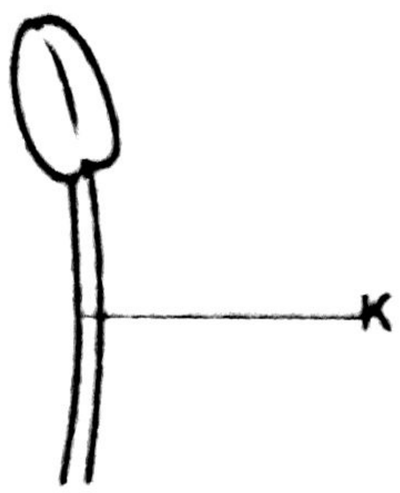
28. Name the non-metallic conductor of electricity found in a dry cell.

29. How does terracing control soil erosion in hilly areas?

30. Give **one** reason why a dog barks when a stone is thrown at it.

31. State **one** way of managing fire outbreaks in schools.

The diagram below shows part of a flower. *Study and use it to answer questions 32 and 33*



32. Name the part of a flower shown in the diagram above

33. Which type of plant gametes are produced by the flower shown in the diagram above?

34. Why is it not advisable to use fresh materials when mulching the garden?

35. How can heat transfer by conduction be put into practice at home?

36. What proves that light energy travels faster than sound energy on a rainy day?

37. Name the class of food whose chemical digestion begins in the duodenum.

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38. How does friction help in the control of motor accidents?

_____

39. How are bacteria useful in the formation of biogas?

_____

40. State **one** importance of starting a Young Farmer's Club in a school.

_____

SECTION B: 60 MARKS

*Questions **41** to **55** carry **four** marks each*

41. a) Give the function of each of the following parts of the eye:

i) retina _____

ii) iris _____

b) State any **one** way the pinhole camera is similar to the human eye in the way they function.

_____

c) Name the eye disease caused by poor feeding.

_____

42. a) Write any **two** factors considered when selecting a good site for setting a hive.

i) _____

ii) _____

b) Give any **one** way:

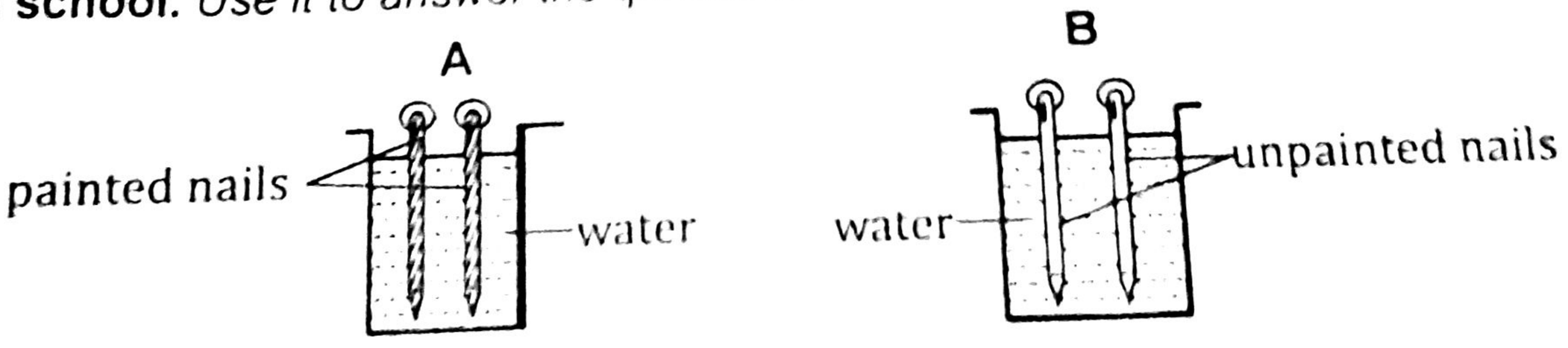
i) bees benefit from a coffee plantation nearby.

_____

ii) coffee plants benefit from an apiary nearby.

_____

43. The diagram below shows an experiment carried out by the science club in a school. *Use it to answer the questions that follow*



a) What was the experiment about?

_____

b) How is painting important to the nails in setup **A**?

_____

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c) Which type of change did the nails in setup **B** undergo after a night?

_____

d) Apart from water, what other condition enabled nails in setup **B** undergo the above mentioned change?

_____

44. a) State the importance of each of the following First Aid Kit in administration of First Aid:

i) splints _____

ii) safety pin _____

b) State the difference between a burn and a scald.

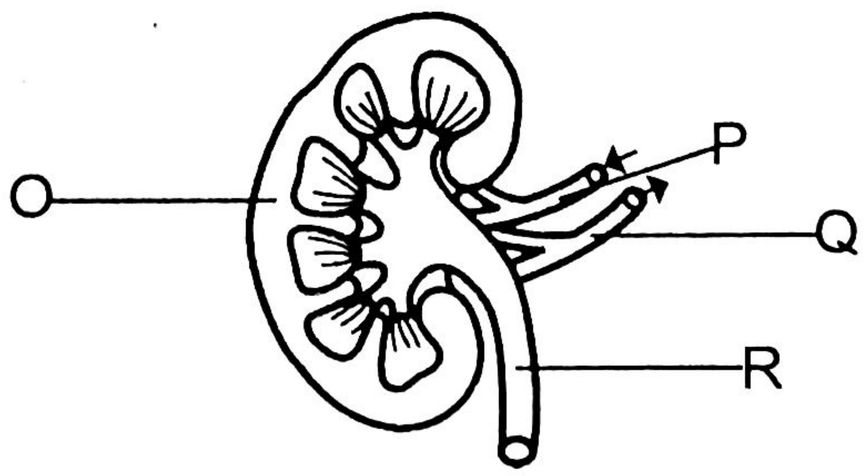
_____

_____

c) Give **one** way you can help a friend with a burn at home.

_____

45. The diagram below shows a human kidney. *Use it to answer the questions that follow.*



a) Name parts labelled **P** and **Q**.

i) **P** _____

ii) **Q** _____

b) What important process takes place in part labelled **O**?

_____

c) To which body organ does part **R** lead **N**?

_____

46. a) What term refers to the scattering of seeds from a parent plant to a well colonised area?

_____

b) Write any **two** adaptations of seeds to being scattered by animals.

i) _____

ii) _____

c) Name any other agent that scatters seeds, apart from animals.

_____

47. a) How does each of the following animals protect itself against enemies?

(i) tortoise _____

(ii) wasp _____

b) Which characteristic of living things is proven when a rabbit runs away from a dog for protection?

_____

c) Apart from the one mentioned in (b) above, state **one** other characteristic of living things.

_____

48. a) Give any **one** example of wood fuel commonly used in our environment.

_____

b) State any **two** ways of conserving energy from wood fuel.

i) _____

ii) _____

c) Why are plants called renewable resources?

_____

49. **Match the simple machines in list A to their groups in list B correctly.**

A	B
screw driver	levers
beam balance	wedges
staircase	wheel and axles
machete	inclined planes

i) screw driver _____

ii) beam balance _____

iii) staircase _____

iv) machete _____

52. a) Name the kind of feeding in babies **that helps to boost the body** immunity

_____

b) Mention any **two** reasons **why babies should be weaned**.

i) _____

ii) _____

c) Why should babies be fed on **soft foodstuffs** during the weaning period?

_____

53. a) State **one** farm practice **that helps to control external parasites** in cattle.

_____

b) What scientific name is given to **chemicals used** to control external parasites in cattle?

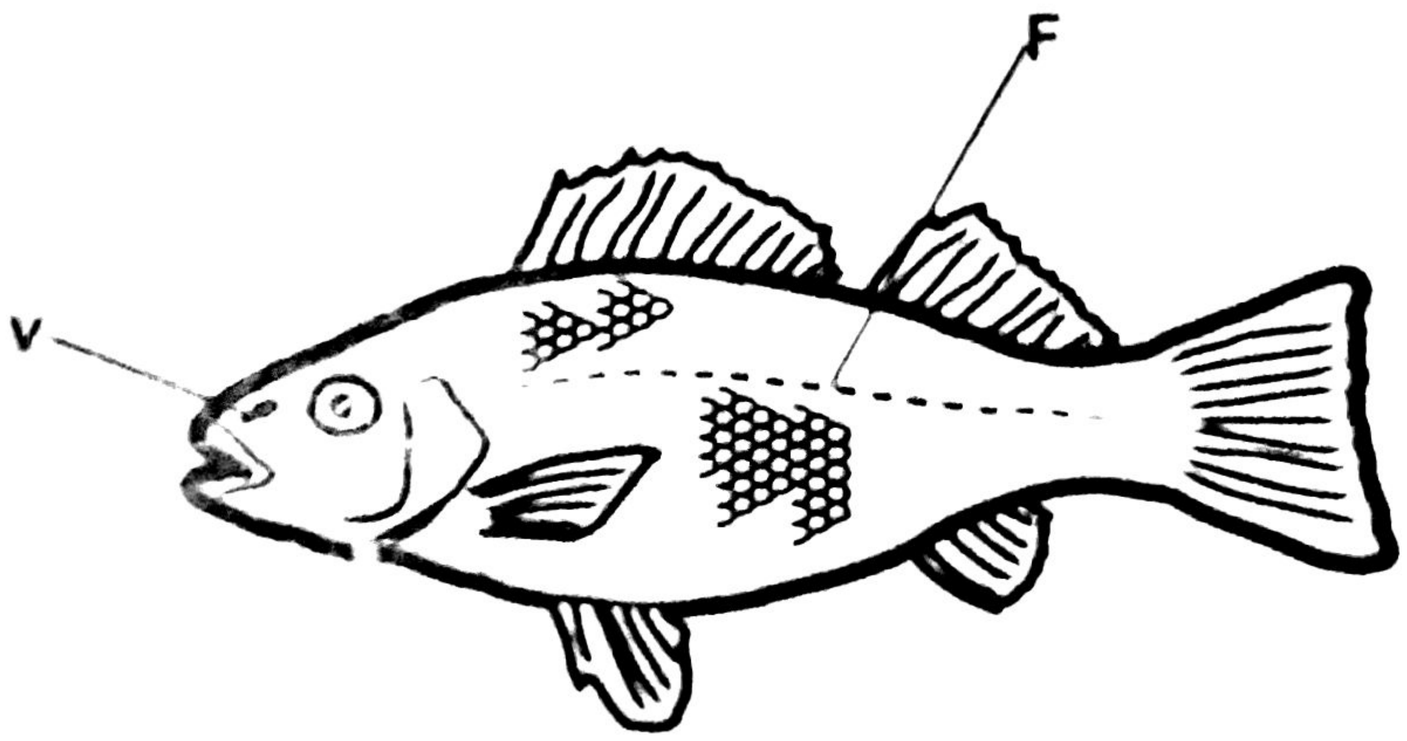
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c) Give **two** effects of external parasites to livestock.

i) _____

ii) _____

54. The diagram below is of a common animal. *Use it to answer the questions that follow.*



a) Name the part labelled with letter **V**.

_____

b) How is part **F** on the animal shown above simillar to the ears on human beings in the way they function?

_____

c) To which group of vertebrates does the animal shown in the diagram above belong?

_____

d) How is fertilization in the animal above different from that in a chameleon?

_____

_____

55. a) Why does photosynthesis take place during day time but not at night?

_____

_____

b) Write the **two** raw materials needed for photosynthesis to occur.

i) _____

ii) _____

c) How does photosynthesis support respiration in animals?

_____

50. a) Name the form of energy that stimulates the sense of hearing in animals.

_____

b) Give **two** other importance of the form of energy mentioned in 50 (a) above to people, apart from hearing.

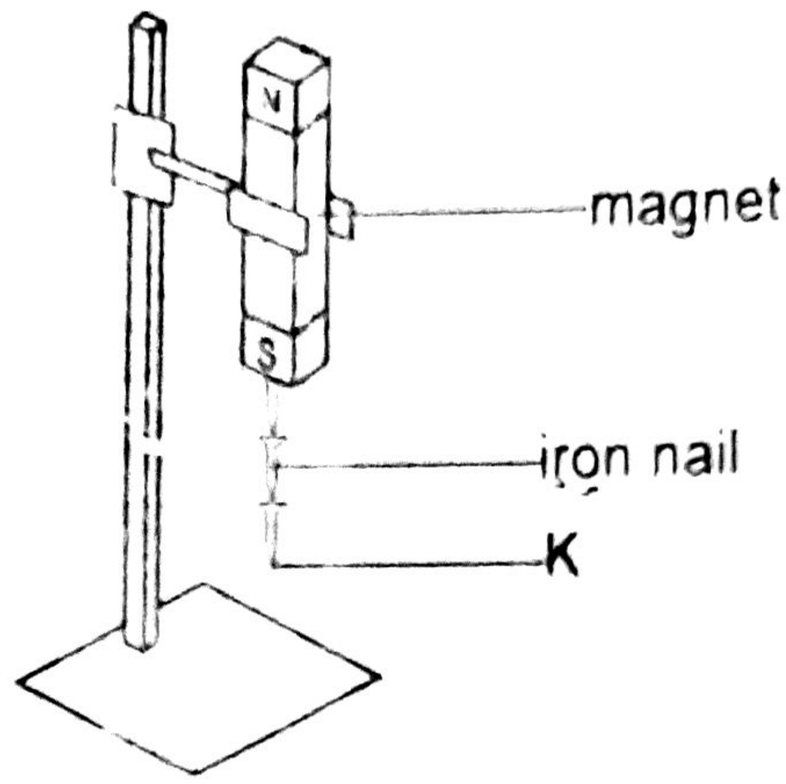
i) _____

ii) _____

c) State **one** way people store the form of energy mentioned in (a) above.

_____

51. **The diagram below shows a method of magnetisation.** *Use it to answer the questions that follow.*



a) How do the iron nails shown in the diagram above become magnetised?

_____

b) Name pole marked with letter K.

_____

c) What scientific name is given to the magnet made using the above method?

_____

d) Give any **one** way of caring for magnets.

_____

CRITICAL THINKING AND EXPERIENCE ACTUAL LEARNING WITH OUR ACTIVITY BOOKS, SEMAS, TEACHER'S GUIDES & PUPIL'S COMPANIONS.